

SMALL-SHAPE EXTERIOR CERTIFICATES FORCE ORIGINAL MIDDLE STATES IN THE ERDŐS–STRAUS EQUATION

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ABSTRACT. For an original exterior state, put $v = a^2/u$, $R = 4a - p$ and $D = (4u + 1)/R$. We retain the integral cubic marked by $\alpha = v - R$, $\beta = v - D$. Every diagonal state $R = D$ has an explicit same-prime middle return with smaller cofactor. At a hard prime, every state with $\max(|\alpha|, |\beta|) \leq 7$ lies on the singular family and forces middle states at its unchanged shell. Radius eight has only one possible nonsingular shape, whose grade congruence also supplies a middle state. This removes these original exterior branches from the remaining E-only search; it does not establish universal ES existence.

An original first-half exterior state at a prime $p \equiv 1 \pmod{8}$ has

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Gcd normalization gives $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$ and $R\kappa = pr + s$. Its ordered denominators are $(a, h\kappa, phr\kappa)$, with inverse $u = a^2/(Ry - pa)$. These Type-I/II coordinates are inherited [1]. Retain

$$v = hs^2 = a^2/u, \quad D = (4u + 1)/R = (r + \kappa)/s, \quad \alpha = v - R, \quad \beta = v - D, \quad B = \alpha\beta - 1.$$

Then $h > 1$, $R, D \equiv 3 \pmod{4}$, $v \neq R, D$, and

$$h \mid B, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}. \quad (1)$$

Primewise quadratic reciprocity gives $(u/R) = (u/p)$, hence $(h/p) = -1$. For a hard prime, meaning $p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}$, the primes 2, 3, 5, 7 are all residues. No positive square can equal α or β : (1) would make every prime factor of h a residue.

The exact marked cubic is

$$Y^2 = X^3 - (\alpha + \beta)X^2 + BX, \quad (X, Y) = (v, 2a), \quad (2)$$

with inverse $R = X - \alpha$, $D = X - \beta$, $a = Y/2$, $p = 2Y - R$ and $u = (RD - 1)/4$. Its original domain requires positive X , positive even Y , positive $R, D \equiv 3 \pmod{4}$, $Y > R$, and the reconstructed prime condition. Its Weierstrass discriminant is $16B^2((\alpha - \beta)^2 + 4)$. At fixed p the line $2Y = p + X - \alpha$ must also be retained.

Theorem 1 (Diagonal channel return). *If $R = D$, write $(R - 1)/2 = \delta t^2$ with δ squarefree. Then $\delta \equiv 3 \pmod{4}$, $\delta \mid p + 1$, and $\delta < R$. Put $c = (p + 1)/\delta$, $j = (\delta + 1)/4$. The original middle state*

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M, Q_M) = (jc, c + 1, j, j, 1, c, 1, \delta)$$

returns (jc, pj, pj) , with $\min(R_M, Q_M) < R$.

Proof. The equality $R^2 = 4u + 1$ makes u and a even. Thus $R \equiv 7 \pmod{8}$ and $z = (R - 1)/2 \equiv 3 \pmod{4}$. Since $u = z(z + 1) \mid a^2$, every prime of z divides a , so its squarefree part δ divides a and $p + 1$. Now $p = \delta c - 1 = 4jc - c - 1$. Substitution proves the gate and the ordered identity. The first-half upper bound is $c(\delta - 1) > 2$, and $\delta \leq (R - 1)/2$. The original fibre retains positive odd t , reconstructing $R = 2\delta t^2 + 1$, $a = (p + R)/4$, $u = (R^2 - 1)/4$ with its original divisibility. $\square$

Theorem 2 (Radius-seven rigidity). *At a hard prime, every original E state with $\max(|\alpha|, |\beta|) \leq 7$ has*

$$h = 4n^2 - 2, \quad r = n, \quad s = 1, \quad R = D = 4n^2 - 1, \quad \kappa = 4n^2 - n - 1$$

for even $n \geq 2$. Its parameter and distinguished denominator are

$$p = 16n^3 - 4n^2 - 8n + 1, \quad a = n(4n^2 - 2).$$

At that same shell $u_M = n$ and $u'_M = n(4n^2 - 2)^2$ are the two original middle orientations, with ordered return (a, pa, pn) .

Proof. There are 52 ordered profiles with nonzero coordinates in $[-7, 7]$ and $\alpha \equiv \beta \pmod{4}$. If $B \neq 0$, its complete list of possible large prime factors gives only

B	(α, β)	h
11	$(2, 6), (6, 2), (-2, -6), (-6, -2)$	11
-13	$(-6, 2), (2, -6), (-2, 6), (6, -2)$	13
-17	$(-4, 4), (4, -4)$	17
-37	$(-6, 6), (6, -6)$	37.

All other nonzero B have prime factors only 2, 3, 5, 7 and cannot contain the nonresidue grade h . Substituting $X = hs^2, Y = 2hrs$ into (2) gives

$$4r^2 = hs^4 - (\alpha + \beta)s^2 + B/h. \quad (3)$$

For $h = 11$ or 17, the requirement $hs^2 - \alpha \equiv 3 \pmod{4}$ is impossible. For $h = 13$, s is odd. When $\alpha + \beta = 4$, (3) gives $4r^2 \equiv 8 \pmod{16}$. When the sum is -4 , it makes r even and then $p \equiv 5 \pmod{8}$. For $h = 37$, (3) makes r, s odd; $p \equiv 1 \pmod{8}$ selects $\alpha = -6$. Modulo three, (3) forces $3 \nmid s$, $3 \mid r$, and hence $p \equiv 2 \pmod{3}$. All nonsingular cases are excluded.

The singular alternatives have $\alpha\beta = 1$. The positive-square obstruction excludes $(1, 1)$. For $(-1, -1)$, $s^2(v + 2) = 4r^2$ implies $s \mid 2$; $s = 2$ fails modulo four. Hence $s = 1$, $h = v = 4r^2 - 2$, giving the stated family. Finally $n + a = n(h + 1) = nR$ proves the original M gate; normalization gives $(h_M, r_M, s_M, \lambda_M) = (n, 1, h, 1)$. $\square$

The family already appears in the preceding exterior cutoff [3]; the result above classifies all the stated hard-prime shapes and supplies its middle return. It also has an endpoint E state with $R_0 = 4n - 1$, $a_0 = n(h - n + 1) < a$, $u_0 = a_0$ and $\kappa_0 = h$.

Theorem 3 (Radius eight has a terminal middle return). *Every hard-prime E state of shape radius at most eight forces an original M state. Beyond the singular case, the only possible shape is*

$$(\alpha, \beta, h) = (8, -4, 11), \quad 4r^2 = 11s^4 - 4s^2 - 3, \quad p = 44rs - 11s^2 + 8.$$

Any such hard candidate satisfies $p \bmod 840 \in \{1, 121\}$. It has the middle return

$$s_M = (p + 3)/11, \quad (h_M, r_M, s_M, \lambda_M) = (1, 3, s_M, 1),$$

with $a_M = 3(p + 3)/11$, $u_M = 9$, $R_M = (p + 36)/11$, $Q_M = 11$.

Proof. Only profiles containing ± 8 need be added. A positive coordinate 4 is excluded. For $B = -65$, the possible grades 13 and 65 fail the residual condition modulo four; $B = 63$ has no large prime. For $(-8, -4)$ or its reverse, $h = 31$ and (3) gives $4r^2 \equiv 12 \pmod{16}$. For $B = -33$, grade 33 fails modulo four; at grade 11, (3) and $p \equiv 1 \pmod{8}$ leave exactly $(8, -4)$. Modulo five the quartic and the hard residue condition give $p \equiv 1 \pmod{5}$. Modulo seven it gives $r^2 \equiv s^4 - s^2 + 1$; the square classes leave only $p \equiv 1, 2 \pmod{7}$. Together with $p \equiv 1 \pmod{24}$ this gives the two stated classes modulo 840. Now (1) gives $p \equiv 8 \pmod{11}$. Thus s_M is integral and $p = 11s_M - 3 = 12s_M - (3 + s_M)$. Primality gives $\gcd(3, s_M) = 1$; the source range follows from $s_M \equiv 4 \pmod{8}$, and the ordered M identity follows by direct substitution. Solving the quartic is unnecessary for this positive return. $\square$

For example, the earlier hard-prime E state $(p, a, u, R, D) = (48049, 12090, 24180, 311, 311)$ maps to

$$\frac{4}{48049} = \frac{1}{12090} + \frac{1}{580912410} + \frac{1}{1873911},$$

with middle cofactor 155. Its exterior factors exceed the earlier middle bound; the actual channel map, rather than an identification of the two state types, repairs that comparison on its exact domain.

The cubic has a rational involution $(X, Y) \mapsto (B/X, BY/X^2)$ when $B \neq 0$. An original positive integral image requires $s \mid 2$. At $s = 2$ its new Y is odd. At $s = 1$ its new residuals are $4r^2 - D$ and $4r^2 - R$, both $1 \pmod{4}$; its new parameter is $3 \pmod{4}$ and differs from p by $(4r - 1)(4r^2 - R - D)$. It is not a fixed-prime descent.

Scope. These are actual same-prime returns from specified original branches. They do not prove that every prescribed prime supplies such a branch or any E/M state. The complete positive-coefficient existence assertion remains unresolved. The supplied standard-library programs retain original factors, exponent vectors, orientations, coefficient fibres and the finite 52-profile proof. No historical priority, independent human mathematical review or Lean build is claimed.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). *Counting the number of solutions to the Erdős–Straus equation on unit fractions*. J. Australian Math. Soc., 94, 50–105. [arXiv:1107.1010](#), §2.
- [2] Hatcher, A. (2022). *Topology of Numbers*. Author edition, §6.4, pp. 205–212.
- [3] The Clankers. (2026). *Exterior three-parameter bound and complete original atlas*. Supplied `exterior_atlas/core.tex`, §§2–5.